

A Clinical Decision Support System for Prescription Risk Screening using Graph-LLM Integration

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Outline

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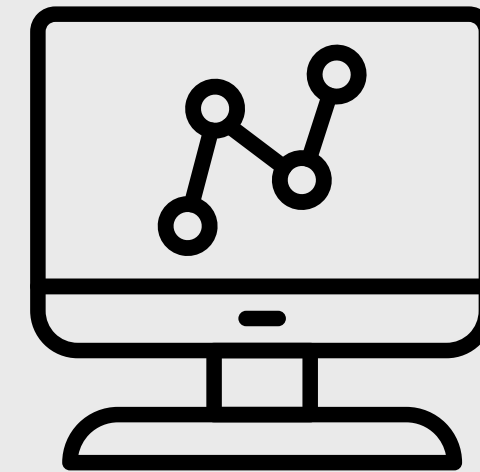
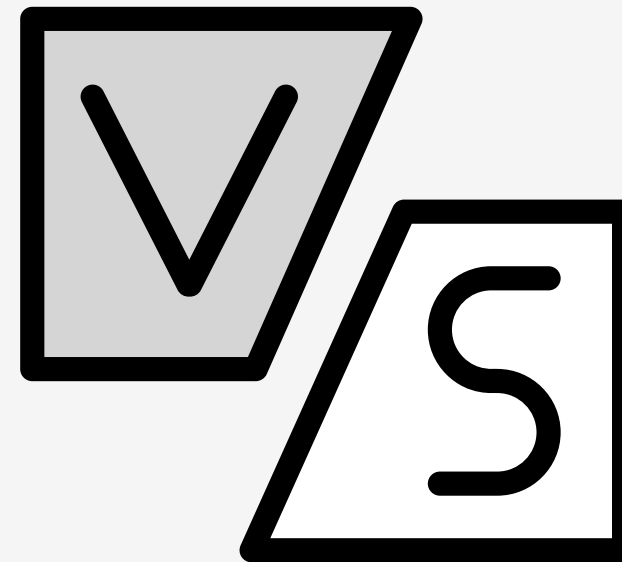
Problem Statement



Complex
cardiovascular cases

Unstructured notes &
indirect risks

Polypharmacy



Current Clinical Decision
Support System

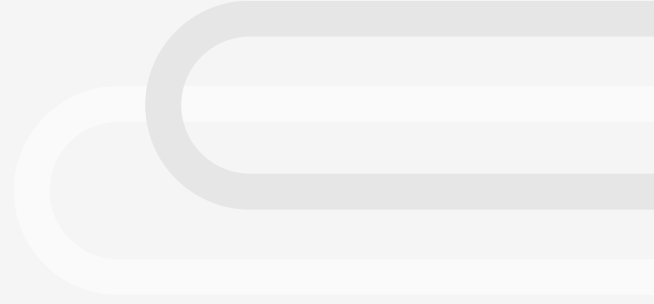
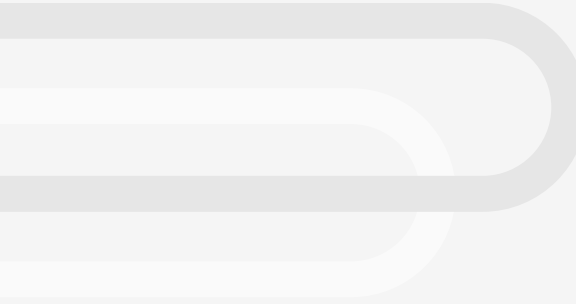
Single-hop name
lookups only

Static

Aims and Objectives

Aim: Build a prototype using an LLM parser + Neo4j knowledge graph for risk screening.

- 01.** Parse clinical notes into structured parameters
- 02.** Build a pharmacological Knowledge Graph (Neo4j)
- 03.** Calculate MERIS risk score automatically



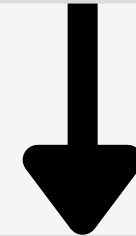
Literature Review

Situating the problem within existing work

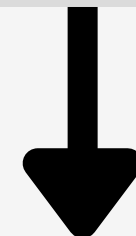


Clinical Context & Limitations

The Clinical Reality: High preventable medication harm in cardiovascular polypharmacy [Lam et al., 2026].



The System Flaw: Current CDSS rely on static table lookups; blind to free-text notes.



The Consequence: Severe "alert fatigue" and undetected indirect risks.

LLMs in Medical Text

Strengths

- High accuracy in extracting features from notes
[Abumelha et al., 2025]

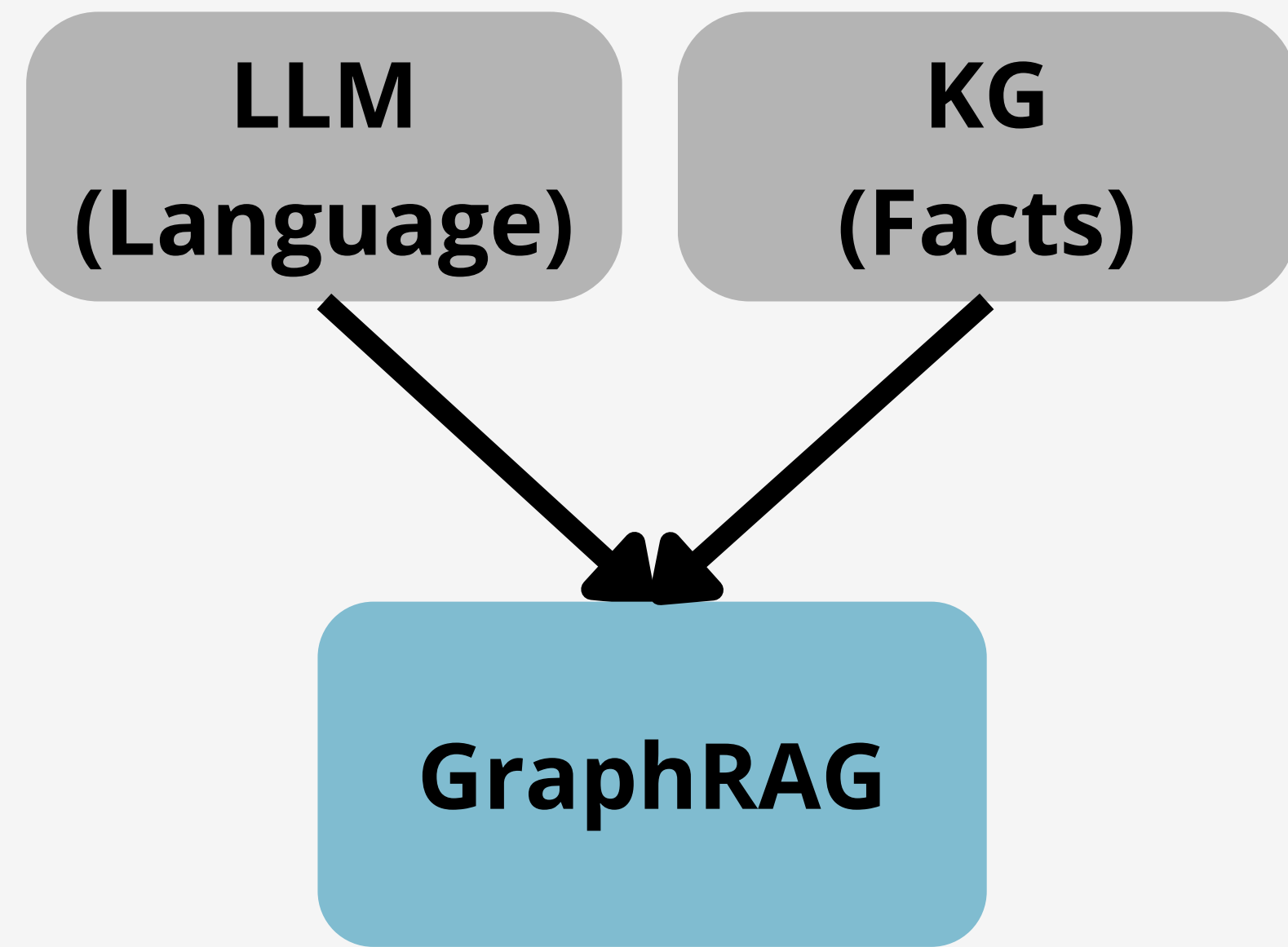
Risks

- Fails at strict logical inference
[Gopalakrishnan et al., 2024]
- Hallucination [Pal et al., 2023]

Conclusion: Cannot be trusted for factual medical decisions

The Gap & The GraphRAG Solution

- Knowledge Graphs: Traverses multi-hop paths to find indirect risks [Lin et al., 2023].
- GraphRAG Framework: Feeds verified subgraphs into the LLM as context [Edge et al., 2025].
- Validation: Compact LLMs + Neo4j achieved ~99.9% retrieval accuracy [Nygren et al., 2026].

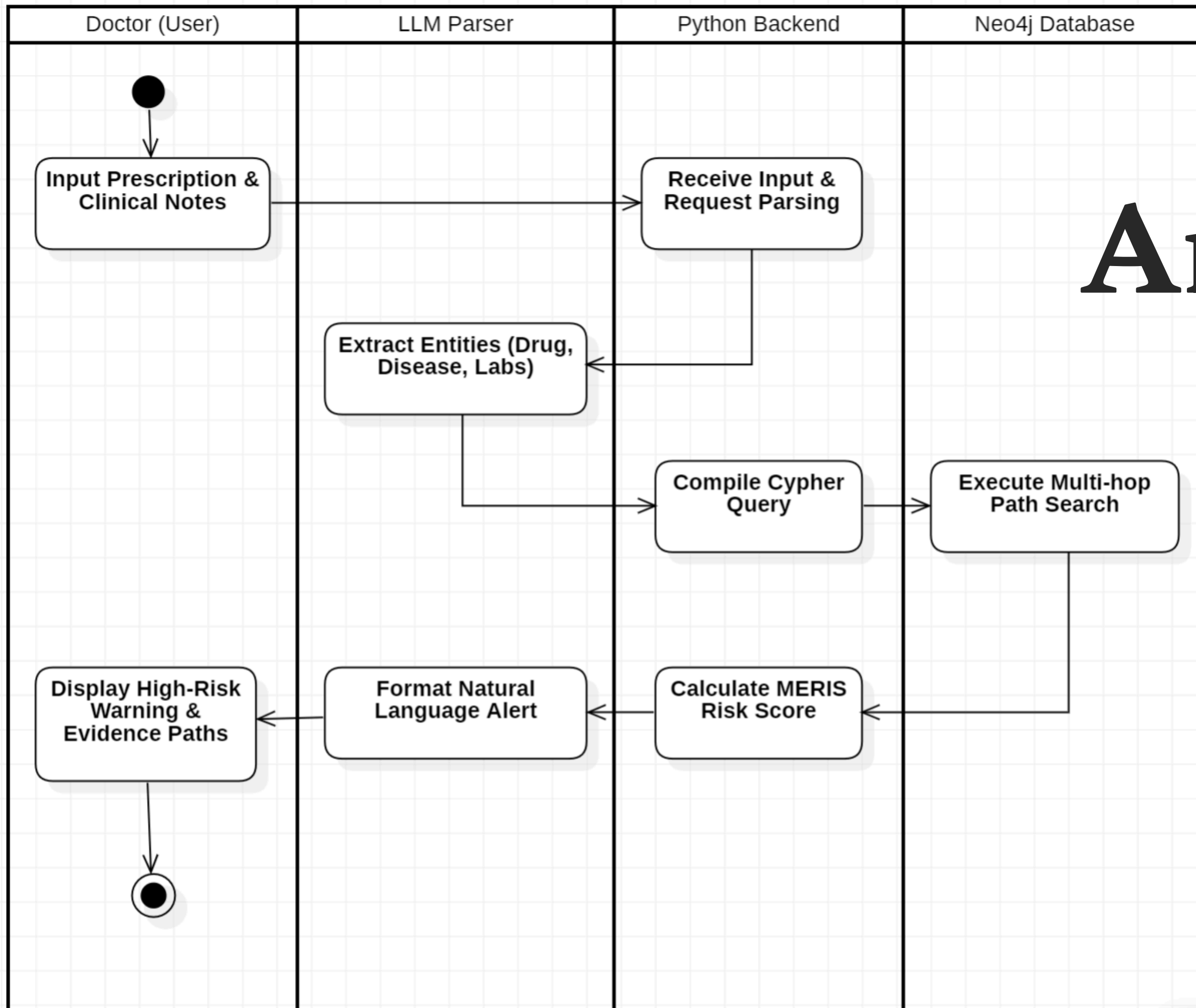


MERIS Framework & Research Gap

- Risk Scoring: Validated MERIS framework.
- Our Goal: Close all three gaps simultaneously.

	Unstructured Text Parsing	Multi-hop Reasoning	Automated MERIS Scoring
Standalone LLM	✓	×	×
Traditional KG	×	✓	×
Existing Scoring (e.g., AUGMERIS)	×	×	✓
Our Proposed System	✓	✓	✓

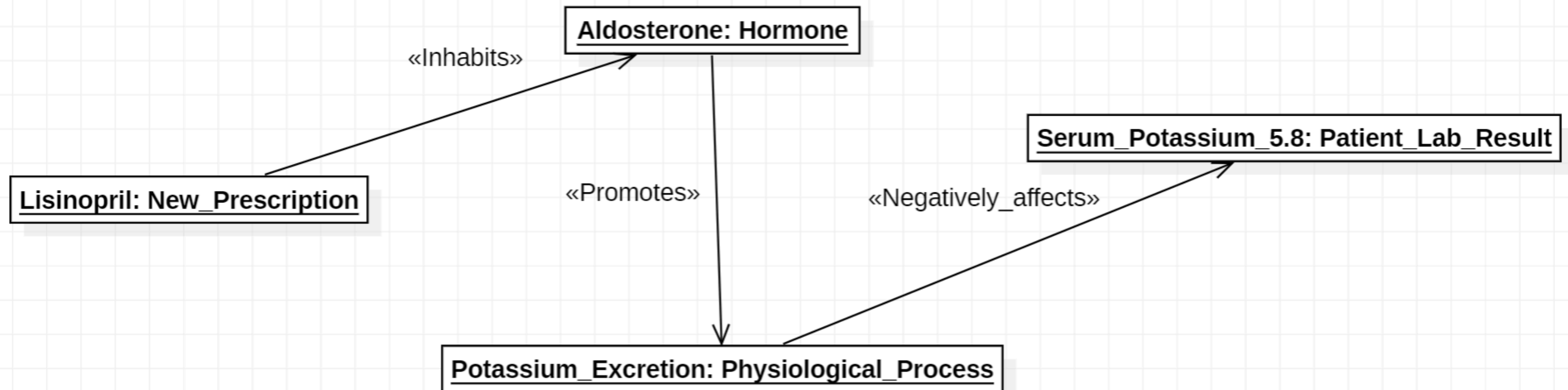
System Architecture



3-Layer Pipeline:

- 1. Parsing (LLM): Text → Variables (No medical judgment).
- 2. Reasoning (Neo4j): Multi-hop risk search.
- 3. Output (Python+LLM): MERIS score → Alert.

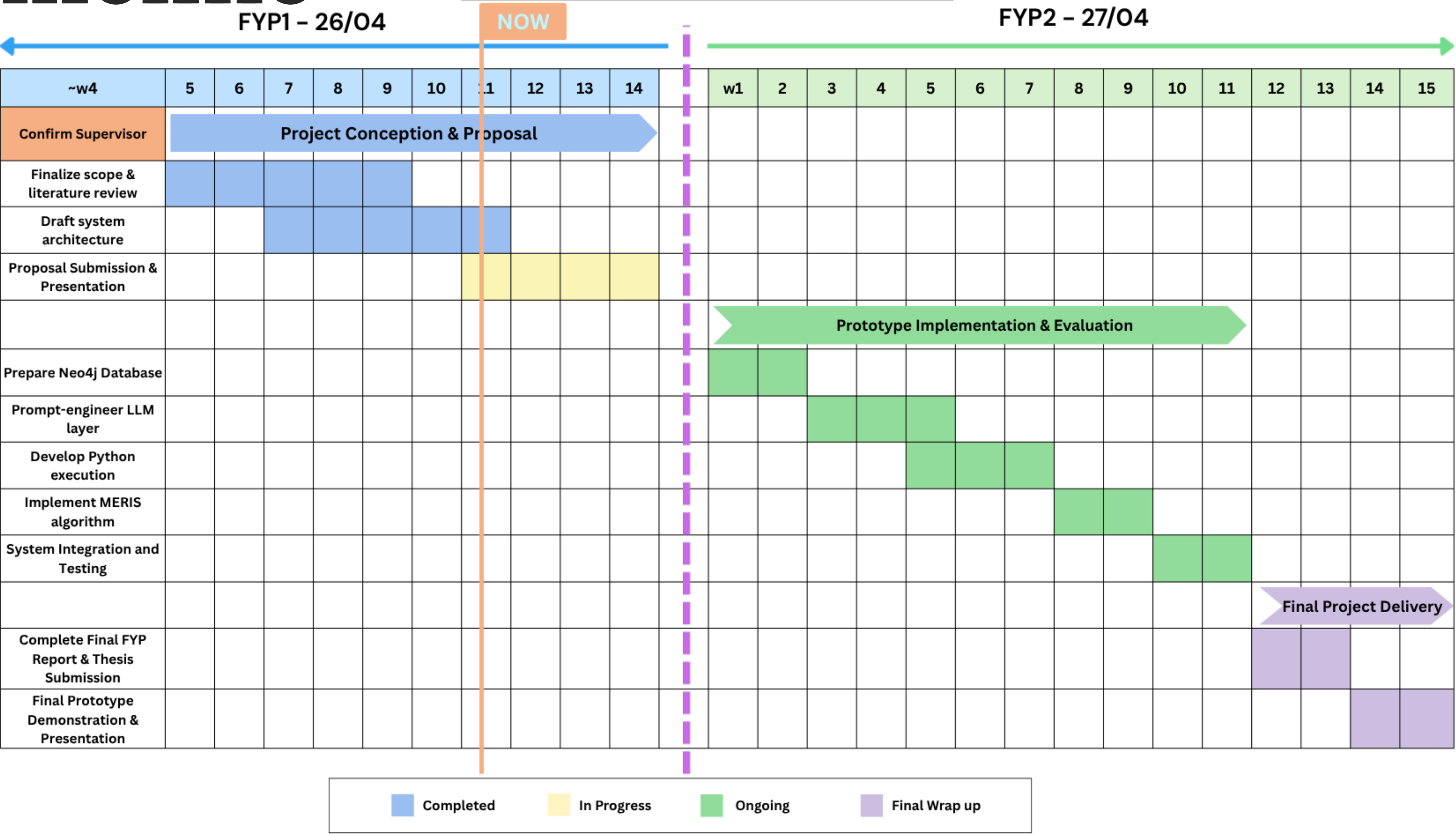
Research Methodology - Multi-hop Reasoning Case



Conflict Detected (Patient-Aware)
Logic: Lisinopril inhibits Aldosterone, which reduces Potassium excretion. This is a severe risk for this patient who already has high serum potassium (5.8 mEq/L).
Calculated Risk Score: 0.92 (Extreme Risk - Hyperkalemia)

Timeline

FYP Project Timeline



Potential Project Significance

Clinical Safety:

Eliminates hallucination in medical decision-making by anchoring AI to explicit graph edges.

Auditable AI:

Provides a transparent, fully traceable reasoning path for regulatory compliance in healthcare AI.

Expected Outcomes & Concluding Remarks

Expected Deliverables:

- 1. Functional Software Prototype (Neo4j Graph + Local LLM Parser + Web UI).
- 2. Performance Evaluation Report (Accuracy & Hallucination rate benchmarking vs. standalone LLM).



Key References

- Saedder, E. A., et al. (2016). Medication Risk Score Algorithm. BCPT.
- Pal, A., et al. (2023). Med-HALT: Medical Domain Hallucination Test.
- Edge, D., et al. (2025). From Local to Global: A Graph RAG Approach.
- Nygren, S., et al. (2026). RAG-based architectures using compact LLMs. Scientific Reports.



Q & A